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NMAM INSTITUTE OF TECHNOLOGY, NITTE*Off-Campus Centre of Nitte (Deemed to be University)***Second Semester M.Tech (Construction Technology) (CBCS) Degree Examinations**

May - June 2024

22CCT202 – CONSTRUCTION AND CONTRACT MANAGEMENT

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from each Unit.**Unit – I**

	Marks	BT*	CO*	PO*
1. a) Define ethics and Discuss any four ethical principles.	6	L*1	1	1
b) Give at least 4 comparisons between Personal Ethics and Professional Ethics.	4	L2	1	1
c) Prepare a preliminary estimate for large Hydro Electric Power Plant, with following data: i) Capacity augmentation: 2500 MW, cost per MW is Rs 1.75 Cr. ii) Environmental mitigation cost is 3% of total project cost. iii) Resettlement cost is 4% of the project cost. iv) Project all risks and contingencies is 2.5% of capacity augmentation cost. v) Project management fee is 2% of gross. vi) Design, survey, geotechnical etc – 7% of capacity augmentation cost. vii) Escalation cost is 7% of gross. GST is 12% of gross, Labor cess is 1% of gross. Profit is 10% of Capacity augmentation cost .	10	L3	1	3
2. a) Highlight the benefits of Product Life Cycle.	6	L2	1	1
b) Explain the following cost estimates. i) Parametric estimating. ii) Three-Point Estimating	4	L2	1	1
c) Discuss the following with relevant equations. i) Power Sizing method of cost estimating. ii) Cost Indexing.	10	L2	1	2

Unit – II

3. a) Discuss importance of quantity takeoff and explain the need of quantity takeoff in owners and contractors perspectives.	6	L2	2	1
b) Discuss four common 'Cost Estimator Practice Traits'.	4	L2	2	1
c) Total quantity of Earthwork excavation using Excavator is 50000 Cum and needs to be completed in 30 days. It is proposed to deploy excavator working to do the job, working for 8 hours a day. If each excavator can give output of 75 Cum per hour. And have a Crew consists of 1 HSKL + 1 SKL + 2 USKL & Paid at Rs 25000 per month for HSKL, 20000 for SKL and 15000 for USKL – all inclusive. Compute total numbers of Labor Crew / gang. Compute the labor Production rate. Compute the Unit Labor Cost for excavation.	10	L3	2	3
4. a) What is Bid rigging? List out at least five signs of possible bid rigging.	6	L2	2	1
b) Timber Formwork for a Roof Slab of a single storey building needs to be completed in 4 days of 8 hours working each. Total quantity of work is 600 Sq m. Labor Crew Consisting of 2 Skilled worker and 2 Unskilled worker is employed does form work of 1.5 Sq.m per Team Hour. Estimate (1) Total team hours required (2) Man-Hours involved (3) Number of Teams or Crew Gang required (4) Total Work Force employed (5) Unit rate of Work, by considering Skilled worker is paid Rs 1000 per day and Unskilled at Rs 600 per day (6) Compute Hourly Labor Rate for this activity.	14	L3	2	3

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Unit – III

5. a)	Explain three types of unbalanced bidding processes.	6	L2	3	1
b)	Why prequalification of contractor is must? Discuss the prequalification's criteria under following aspects: i) Technical ii) Commercial iii) Ethical	6	L2	3	2
c)	Factors affecting the bid/ no bid decision under following heads: i) Project characteristics: ii) Company characteristics/situation.	8	L2	3	1
6. a)	Highlight the information to be given in tender notice.	4	L2	3	1
b)	Explain the three essential requirements of valid contract	6	L2	3	1
c)	Explain the Termination of Agreements	10	L2	3	3

Unit – IV

7. a)	Discuss any four Construction Contract Provisions to be negotiated with the owner.	08	L2	4	2
b)	Explain with reference to FIDIC, the following i) Liquidated Damages and Bonus for Early Completion ii) Breach of Contract	12	L2	4	1
8. a)	Discuss the advantages and disadvantages of following types of contracts. i) Item rate or unit price contract ii) Cost Plus type Contract.	8	L2	4	2
b)	Discuss in brief the following clauses of a standard set of conditions of contract i) Performance security/ security deposit ii) Suspension of work iii) Time limit for completion	12	L2	4	1

Unit – V

9. a)	Explain the following methods available for alternate dispute resolution: i) Conciliation ii) Dispute Review Boards / Adjudicators	8	L2	5	1
b)	Discuss the six things to do when faced with a dispute in contract management.	12	L2	5	1
10. a)	Discuss the situations leading for extra work and change order, without dispute in Civil Engineering Contracts.	12	L2	5	2
b)	Highlight the highlights of "Arbitration and Conciliation Act 1996".	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
